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Does food recognition depend on color?

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Abstract:

Color is considered important in food perception, but its role in food-specific visual mechanisms is unclear. We explore this question in North American adults. We build on work revealing contributions from general visual abilities as well as those more specific to food, in the recognition of images of prepared food. This prior work also found a negative correlation between the visual abilities specific to food and food neophobia (FN, aversion to novel food). In Study 1, participants performed two food recognition tests, one in color and one in grayscale. Removing color from one of the two food tasks reduced performance, but performance on the food tests with and without color remained correlated, above and beyond any contribution from general visual object recognition ability. We still found a negative correlation between food recognition and FN. In Study 2, we removed color from both food tests. Performance on the food tests with and without color remained correlated, above and beyond the contribution from general visual object recognition ability. However, without color, food-specific recognition ability was no longer related to FN. In Study 3, color-blind men reported lower FN than men with normal color perception. These results suggest two separate food-specific recognition mechanisms, only one of which is dependent on color.

Color has long been considered essential for recognizing food, including the notion that the coloration of plants and the ability of animals to detect it have co-evolved (Polyak, 1957). Color can help to decide whether food is cooked, fruit is ripe or rotten, and people prefer colorful meals (Paakki et al., 2019). Color is arbitrary for many objects but is more constrained for food. Some work suggested that color is not used to recognize food, even when it is diagnostic (Biederman & Ju, 1988). Others found that color facilitate naming fruits and vegetables (Ostergaard & Davidoff, 1985; Wurm et al., 1993). Color plays a central role in some cognitive models of concept representations. One explanation for rare category-specific semantic deficits for fruits and vegetables (Crutch & Warrington, 2003; Samson & Pillon, 2003; see Capitani et al., 2003 for a review of 79 such cases) centers around the unique importance of color in their representation (Warrington & McCarthy, 1987).

The idea that color is critical in the representation of food was recently revisited in a re-analysis of the Natural Scenes Dataset (NSD, Allen et al., 2022). While earlier work concerned judgments about single objects at the basic level (e.g., producing “apple” for a picture of an apple on a white background), this work considers neural responses to complex and varied natural scenes, many of them including food. The NSD includes fMRI measures of brain activity in response to tens of thousands of natural images in eight participants tested over multiple (30 – 40) sessions. Pennock et al. (2022) found brain areas responsive to color in parts of the ventral visual pathway and discovered that these areas were also food-selective. The NSD was however, not collected to address color’s role in the representation of food and the finding amounted to a correlation between selectivity for color and for food. The visual areas of interest responded to color

in the absence of food, but Pennock et al. (2022) suggested that they were specialized for the visual features common to food objects and responded to them even when no food was in the image. Interestingly, two other studies analyzing the same dataset independently also identified similar food-selective areas in the ventral visual pathway, but concluded that low- or mid-level features like color were unlikely to underlie such selectivity (Jain et al., 2022; Khosla et al., 2022). One of these studies identified a component of brain responses using a data-driven approach and found that it was still food-selective when color properties were controlled, but also still related to color when food-selectivity was controlled (Khosla et al., 2022).

It is difficult to assess how critical color is for food-selective responses when food images without color are not included. Several results suggest we should be cautious not to overstate the role of color in food perception. Important decisions about food, such as its healthiness, can be modeled successfully without any reference to color (Gandhi et al., 2022). There was no influence of color when brain regions in the ventral pathway showed responses to food correlated with caloric density (Tang et al., 2014). In other work, even though a food/non-food distinction could be achieved in the occipital cortex 85ms post stimulus onset, neither spatial frequency content nor color helped to classify images as food or non-food (Tsourides et al., 2016). While color is undoubtedly important in the perception of food, it is unclear how necessary it is in engaging food-specific visual mechanisms.

Here, we address how food-specific visual mechanism(s) are supported by the processing of color using an individual differences approach, combined with an experimental manipulation of color. Our approach builds on recent work (Gauthier and

Fiestan, 2022) that characterized a food-specific visual ability and found that it was in part related to food neophobia (FN), defined as the tendency to avoid novel foods (Rabadán & Bernabéu, 2021). FN, often measured using the short Food Neophobia Scale (Pliner & Hobden, 1992), is considered to be a trait with some negative consequence on healthy eating and individual health (Rabadán & Bernabéu, 2021). Gauthier and Fiestan, (2022) introduced two food recognition tests: a 3-alternatives sequential-matching task and a 4-alternatives oddball task. These tests measured the ability to discriminate visually-similar dishes, such as a quiche and a pumpkin pie, without naming. Performance on these two tasks was correlated even after controlling for domain-general object recognition ability (o , Richler et al., 2019). Furthermore, FN was the strongest predictor of food recognition performance, after o was controlled. This correlational result suggests that individuals who avoid novel food may, through relatively limited visual experience, end up with poorer perceptual expertise in this domain. Here, we take advantage of this work and ask whether a food-specific ability can still be captured with grayscale images of food, and if so, whether color-independent food recognition ability is related to FN.

In Study 1, we randomly assigned participants to perform the tasks from Gauthier and Fiestan (2022), but converted the images to grayscale for only one of the two food recognition tests. We test whether recognition of grayscale food images shares a domain-specific component with a color food test, after controlling for o . In Study 2, we replicated Gauthier and Fiestan (2022, this time using grayscale images in both tests. We quantify the proportion of the variance in food recognition tests that is domain-specific when color can play no role and assess whether FN still predicts the recognition of

grayscale food images. In Study 3, we test a prediction stemming from the results of Study 2, which is that color-blind men may show less FN than those who are not color-blind.

Open Practices Statement.

The study was not pre-registered. The materials for the food tasks were made available before (Gauthier & Fiestan, 2022) and the data for all studies are publicly available (Gauthier, 2022).

Study 1

Method

All aspects of the study were identical to those in Gauthier and Fiestan (2022), except that each participant was assigned to a condition in which one of the two food tests was in grayscale (the other task remained in color). This design of course allows us to measure the impact of removing color on food recognition performance for each task. But while we expected that removing this information would reduce average performance, we were even more interested in how removing color for one of the two food tasks would affect the correlation between the two food tasks. Would the two food tests still correlate after controlling for d , when one of them could not measure any variance related to color information? If we can measure a food-specific ability when color is absent from one of the tests, we wanted to know if this ability would correlate with FN, as in the previous work.

Participants.

Participants (18 to 55 years old, fluent in English and with a minimum approval rating of 95% on the platform) were recruited from the US/Canada on Prolific.co. Following Gauthier & Fiestan (2022), we excluded anyone self-reporting with a diagnosed or suspected eating disorder. We used Bayesian analyses, starting with a target sample size of at least 120. Participants were randomly assigned to two groups (only differing in whether the food matching or food oddball task was done in grayscale). Originally, we collected 127 participants (62 in the matching-gray group, 65 in the oddball-gray group). Two participants in each group failed attention checks and were replaced. Some participants did not complete the experiment and were not included (2 in the matching-gray group, 3 in the oddball-gray group) and one participant in the gray-matching group was excluded for below performance (13%) on the color food oddball test (a bivariate outlier in the correlation between the food tests, Cook's distance of .17). In the end, we analyzed data for 57 participants in the gray matching group and 60 in the gray oddball group. The study was approved by the local Institutional Review Board. Participants provided informed consent and were paid \$3.40. Detailed demographic information in supplemental materials.

Design and procedure.

Testing occurred online on a computer. To provide the same context as in Gauthier and Fiestan (2022), we measured all the same survey variables because the measurement of FN and the visual tasks occurred last in that study. Measures were taken in the same order for all participants and we describe these measures, and the zero-order

correlations between all variables, in the supplemental materials. Here, we focus on the visual tasks and FN measurement to address our hypotheses.

Measures.

The study began with measurement of a few variables that are not of interest for the present hypotheses. This included questions about general interest in visual recognition, about interest in food and a personality scale (Donnellan et al., 2006). See supplemental material for details.

We used the food neophobia scale (FNS, Pliner & Hobden, 1992) to measure FN. The FNS includes 10 items (e.g., “I don’t trust new foods”, “I am afraid to eat things I have never had before”, “I will eat almost anything” (reverse coded), rated on a scale from 1: disagree strongly to 7: agree strongly.

As in Gauthier & Fiestan (2022), we collected data in four visual tests. Two of them were related to food and together were used to estimate food recognition ability. The other two were tests using novel objects, and together were used to estimate domain-general object recognition ability (o, Richler et al., 2019). We describe the tests below in the order taken by the participants.

The first visual test was a Food Matching test. The test started with 1 example trial, followed by 60 experimental trials. On each trial, an image of prepared food appeared for 1000 ms, followed by three images of prepared food (Figure 1A and C). Participants chose one image that was “most similar to the studied image” without time restriction. Details, and the list of target foods, including dishes varying in ethnicity and

expected familiarity for our north american target population¹, is provided in Gauthier and Fiestan (2022) and the test is available online (<https://doi.org/10.6084/m9.figshare.19641429.v1>). This test was scored using accuracy with chance at .33. Participants in Group 1 performed this test with grayscale images and those in Group 2 performed it with color images. We note that the images in this test as well as in the other food test described below were not selected systematically in any way with regards to color for the original study by Gauthier and Fiestan (2022). They were selected through internet searches for food items of various categories and the full color versions were used as they were found.

The second visual test was the Novel Objects Memory Test (NOMT, Richler et al., 2017) with one specific category of novel objects that share a common arrangement of parts (symmetrical Greebles, details of the task in Richler et al., 2017). Participants studied six different Greebles shown together on the screen for 20 sec, 3 times during the test, which included 48 trials 3-alternative forced choice trials with 3 Greebles shown on the screen until one was selected as 1 of the 6 studied targets. This test was scored using accuracy with chance at .33.

The third visual test was the food oddball test. Four images of food were presented on each trial, three showing different versions of the same dish and one a different dish that varied in its visual similarity to the others, depending on the trial difficulty. Participants were to choose the oddball without any time restriction, for 60

¹ We acknowledge that this notion of familiarity is one based on intuition and that familiarity may vary greatly across different parts of the country, agglomeration size and the cultural backgrounds of our participants. Because our tasks do not require naming, familiarity is not necessary for our tasks – in the same way that familiarity with specific individuals is not necessary for faces to be included on a measure of face recognition, or familiarity with a specific bird species is not required to perform well on a bird matching task.ea

total trials (Figure 1B and D). This test was scored using accuracy with chance at .25.

The list of the food categories, including dishes varying in ethnicity and expected familiarity for our target population, and other details are in Gauthier and Fiestan (2022) and the test is available online (<https://doi.org/10.6084/m9.figshare.19641423.v1>). Participants in Group 1 performed this test with color images, and those in Group 2 performed it with grayscale images.

The final visual test was the novel objects matching test (Object matching, adapted from Sunday et al., 2018) followed and included 4 practice trials (with correct/incorrect feedback) and 70 experimental trials (without feedback). Following a fixation cross shown for 500 ms, one image of a novel object appeared for 300 ms, followed by a category-specific mask for 300 ms, and then a second image with “same” and “different” buttons. Participants judged whether the second object was the same as the first, independent of size or viewpoint. There was a 4 sec deadline to respond, and the answer scored as incorrect if no response was provided. The novel objects were from 4 different artificial categories (asymmetrical Greebles, vertical Ziggerins, Horizontal Ziggerins and Sheinbugs), with each category in a distinct color (but color was never diagnostic). There were 31 same trials and 39 different trials. This test was scored using

sensitivity (d' , chance = 0), adjusting for hit or false alarm rates of 1 to $(2n - 1)/2n$ and 0 to $1/2n$ where n is the number of same or different trials, respectively.

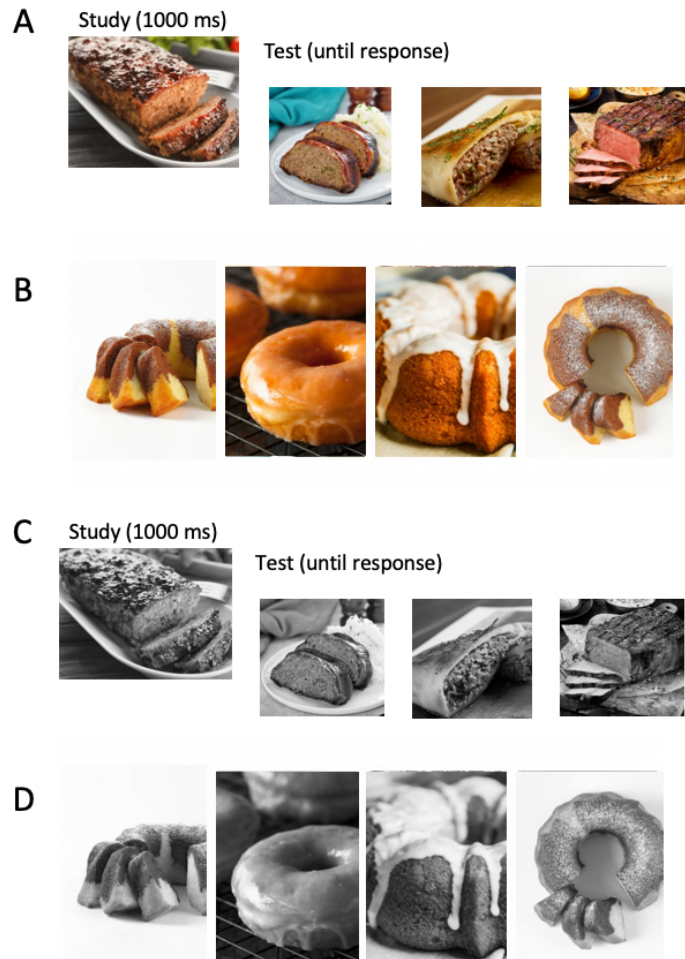


Figure 1. Examples of trials similar to those on the food matching test (A, C) and the food oddball test (B, D), in color (A, B) or in grayscale (C, D), using royalty-free images. The correct answers are the first option in A and C and second option in B and D.

Analyses.

To assess differences between conditions, we used Bayesian estimation, a probabilistic approach allowing quantification of parameter estimates and uncertainty in

the form of the posterior distribution, and aligned with the goals of so-called *new statistics* (Cumming, 2014; Kruschke & Liddell, 2018). For comparisons of means, we used the online implementation of the Bayesian Estimation Supersedes the *t*-test (BEST) model (Bååth, 2012). For correlations, we used JASP (JASP Team, 2022) with a default prior of stretched beta width of .3. To test for differences (between two means or between a parameter and 0), we reported the 95% credibility interval (95% CDI). A difference is deemed credible when the 95% CDI does not include 0. When 0 is included in the 95% CDI, we also provide the BF_{01} to quantify support for the null hypothesis (for BF_{01} of means comparisons, a default prior of Cauchy scale .707 was used). We used Jeffrey's interpretation rules (Jeffreys, 1961), with values between 1 and 3 deemed inconclusive. To provide information on sensitivity to priors (Depaoli et al., 2020), we also report the smallest prior value that renders a decision inconclusive.

Results.

Table 1 provides descriptive statistics for all measures. Aggregate scores for *o* and food recognition ability were computed as in Gauthier and Fiestan (2022), using the average of the Z-scores for each test. Most importantly, reliability (Guttman's λ_2 , Tang & Cui, 2012, or weighted reliability, for the aggregate scores, Wang & Stanley, 1970) was acceptable for high for all variables (.75-.91). The FNS scores are similar to those in a study of 1398 college students in the US (FNS = 29.8, SD=11.7, Olabi et al., 2009).

Table 1 Descriptive statistics for individual tasks and aggregates

	Mean	SD	λ^2	Skewness	Kurtosis
<i>Group 1 Food Matching Gray</i>					
NOMT-SymGreeb (acc)	0.54	0.17	.86	-0.21	-0.51
Object matching (d')	0.98	0.72	.82	0.10	0.44
Food matching (acc)	0.69	0.10	.76	-0.28	-0.08
Food oddball (acc)	0.66	0.13	.84	-0.43	-0.06
Food ability (z-score)	0	0.86	.86	-0.15	-0.11
<i>o</i> (z-score)	0	0.82	.88	-0.02	-0.47
FNS	29.12	11.60	.91	0.14	-1.07
<i>Group 2 Food Oddball Gray</i>					
NOMT-SymGreeb (acc)	0.55	0.15	.81	0.34	0.16
Object matching (d')	1.09	0.65	.76	0.23	0.09
Food matching (acc)	0.82	0.10	.83	-1.22	1.34
Food oddball (acc)	0.56	0.11	.75	-0.37	-0.60
Food ability (z-score)	0	0.85	.87	-0.70	-0.21
<i>o</i> (z-score)	0	0.75	.81	-0.29	-0.43
FNS	28.13	10.40	.89	0.50	0.00

We first addressed the impact of performing the food tests with grayscale images (Figure 2) –we expected that removing color would increase difficulty and simply assess this as a manipulation check, to show that color was informative in these tests. Each test had a different chance level and there was no interest in comparing across tests, which were not matched in difficulty. We compared performance across the two groups for each test. Participants were randomly assigned to a group and all other aspects of the study were identical aside from one food task using grayscale images. As expected, the 95% CDI for the mean difference between groups for the NOMT overlapped with 0 (-.070,

0.048; Cohen's $d^2 = -.058$; $BF_{01} = 4.749$, moderate or stronger evidence obtained for all priors above .41). Similarly, the 95% CDI for the mean difference between groups for the Object matching overlapped with 0 (-0.359, 0.144; Cohen's $d = .16$; $BF_{01} = 3.552$, moderate or stronger evidence obtained for all priors above .58). In contrast, we found very strong evidence for group differences when the presence of color was manipulated. For the Food matching test, the 95% CDI was well above 0 (0.089, 0.166; Cohen's $d = 1.34$), as well as for the Food oddball test (0.063, 0.149; Cohen's $d = 0.91$). In fact, in both cases 100% of the posterior distribution fell above 0. Despite the large effect sizes on food tests (a 15% and 16% reduction in performance relative the color versions), all participants performed above chance in grayscale.

² BEST model returns effect size computed as $(\mu_1 - \mu_2) / \sqrt{(\sigma_1^2 + \sigma_2^2) / 2}$ (Kruschke, 2013). It uses estimated parameters of posterior distribution, unlike Cohen's d which is computed by mean and standard deviation for the data. For simplicity, we refer to the effect size from BEST model as Cohen's d .

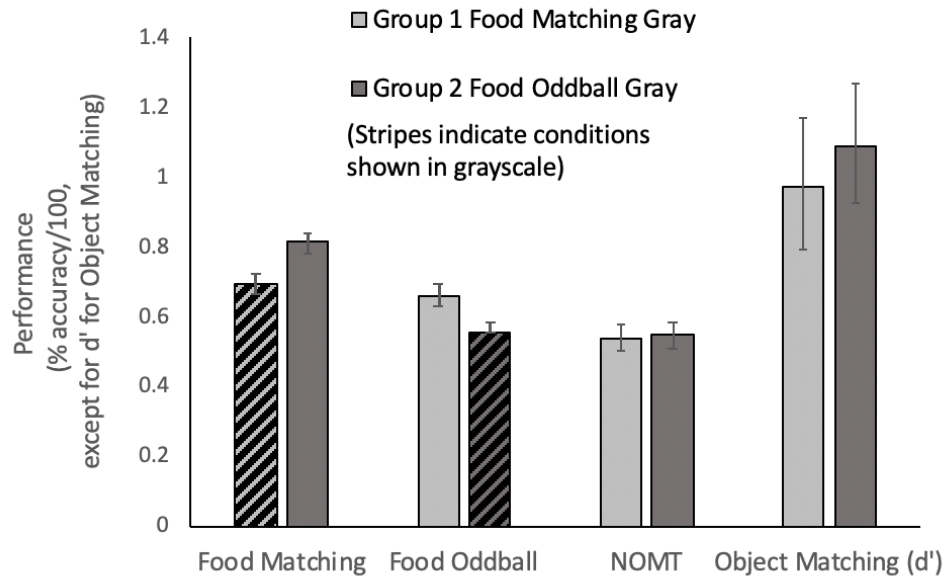


Figure 2. Mean performance in each visual test in Group 1 and Group 2 for Study 1. Note that object matching was scored as sensitivity and should not be compared with the first 3 tests. Error bars show the 95% CIs.

Next, we investigated the relationships between individual differences in our main variables within each group. The two groups provided a near replication of the same manipulation (removal of color in one of the food tests) for these analyses. We used a directional stretched beta prior width of 0.3, which reflects the belief that the correlations will be in the same direction as in Gauthier and Fiestan (2022), and that moderate effect sizes are more likely than very large effect sizes.

We found credible evidence that Food ability and o were positively correlated in both groups (Group 1: $r = .35$, 95% CDI: .084, .522; Group 2: $r = .41$, 95% CDI: .148, .566, see Figure 3). We also found credible evidence that the two food tests were positively correlated after controlling for o , in both groups (Group 1: $r_{\text{partial}} = .41$, 95% CDI: .143, .571; Group 2: $r_{\text{partial}} = .37$, 95% CDI: .113, .539, see Figure 3). This supports the idea that a food-specific ability can be measured regardless of the presence of color.

Finally, we found credible evidence that Food ability residuals (controlling for o) was correlated negatively with FNS (Group 1: $r_{\text{semi-partial}} = -.26$, 95% CDI: $-.456, -.034$; Group 2: $r_{\text{semi-partial}} = -.27$, 95% CDI: $-.461, -.041$).

Removing color from one of the food tests does not abolish evidence for the main results in Gauthier and Fiestan (2022), suggesting that a great deal of what was measured in the original work was independent of color. At least qualitatively, the most important influence of removing color for one of the food tests was a reduction of the variance shared between tests (from $r = .55$ in Gauthier & Fiestan (2022) to $r_s = .41$ and $.37$). Gauthier and Fiestan (2022) may have underestimated the domain-general variance contributing to food recognition performance because part of it was color specific. Removing the color from one of the two food tests takes away the possibility of any domain-general or domain-specific variability due to color contributing to the shared variance between the tasks.

The still robust correlation between food-specific recognition ability and FNS is more difficult to interpret because the aggregate food ability includes a mixture of abilities measured with grayscale and color. The correlation between food-specific ability and FNS is a replication of the original result with color images of food in both tests, but it is unclear whether it means that this relationship is independent of color.

To address this question, we conducted Study 2, identical to Study 1 in all respects except that this time, instead of removing color for only one of the food tests, participants performed both food tests with grayscale images. We expect an even stronger correlation between o and food recognition ability because none of the variance in performance on the food tasks can be driven by color.

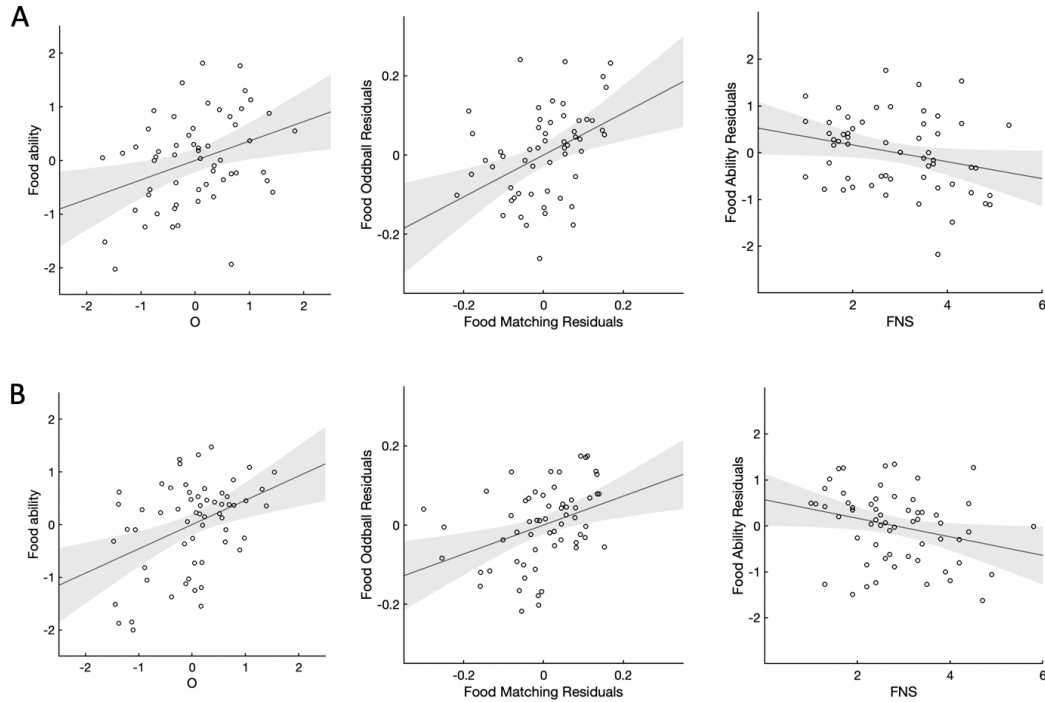


Figure 3. Scatterplots showing the zero-order correlation between Food ability and O , the partial correlation between the two food tests controlling for O , and the semi-partial correlation of food ability residuals and FNS, for Groups 1 (A) and Group 2 (B) in Study 1. Shaded areas show the 95% C.I.

Study 2

Method.

Participants.

The same website and criteria were used as in Study 1. We started with a sample size of 60 and added 20 additional participants until we had decisive support for or against the correlation between FNS and food-specific recognition. Two participants' data were removed because their performance on the food oddball test was below chance (they were also both bivariate outliers, with Cook's distance $>.3$). One additional

participant's data were removed because they failed attention checks. The study was approved by the local IRB. Participants provided informed consent and were paid \$3.40. We ended collection after seventy-seven participants (demographic details in supplementary materials).

Design and procedure.

All aspects of the study were identical to Study 1, except that all participants performed the two food tasks in grayscale versions.

Results.

Table 2 presents descriptive statistics. Performance on the food tasks was comparable to grayscale versions in Study 1.

Table 2. Descriptive statistics for individual tasks and aggregates

	Mean	SD	λ^2	Skewness	Kurtosis
NOMT-SymGreeb (acc)	0.54	0.13	.75	0.49	-0.16
Object matching (d')	1.16	0.68	.77	-0.37	-0.43
Food matching (acc)	0.71	0.09	.66	-0.34	-0.31
Food oddball (acc)	0.55	0.11	.76	-0.10	0.10
Food ability (z-score)	0	0.89	.82	-0.04	-0.37
o (z-score)	0	0.80	.81	-0.07	-0.48
FNS	29.20	12.69	.92	0.50	-0.81

We investigated individual differences between our main variables (for other survey variables, see Supplementary materials). We found credible evidence that Food ability and o were positively correlated ($r = .57$, 95% CDI: .383, .694, see Figure 4). This is consistent with Study 1, suggesting that food-recognition depends at least in part on the ability to process shape information. We also found credible evidence that the two food tasks were positively correlated after controlling for o ($r_{\text{partial}} = .44$, 95% CDI: .211, .574, see Figure 4). This is also consistent with Study 1 in suggesting that the food-specific ability is not entirely dependent on color. For the correlation between Food ability residuals and FNS, the 95% CDI did not technically overlap with 0 (at 3 decimals points), but the effect size was very small and was not negative ($r_{\text{semi-partial}} = .02$, 95% CDI: -.230, -.003). We found evidence against a negative correlation ($\text{BF}_{-0} = .238$, this conclusion

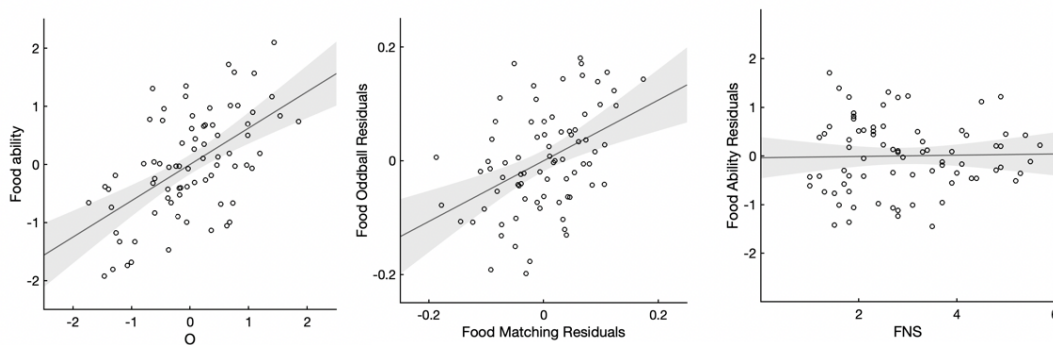


Figure 4. Scatterplots showing the zero-order correlation between Food ability and o , the partial correlation between the two food tests controlling for o , and the semi-partial correlation of food ability residuals and FNS in Study 2. Shaded areas show the 95% C.I.

holds for all priors above .16) and we also found evidence in support of a null correlation ($\text{BF}_{01} = 3.583$, with moderate support for the null for all priors above .21). The correlation between food-specific recognition ability and FNS, observed in Gauthier & Fiestan (2022) and in Study 1, does not replicate in the absence of color.

This suggests that one component of food recognition ability is completely independent from color, and another component relating to FNS may be entirely dependent on color. One prediction based on these results is that individuals with color perception abnormality could experience less food neophobia because of a reduced range in the feature that we found most relevant to this trait. In Study 3, we tested this prediction.

Study 3

Method.

Participants.

Color blindness is a decreased ability to perceive some differences in color, which can impact judgments about food (e.g., judging ripeness, detecting mold, deciding when meat is cooked). It affects approximately 8% of Caucasian males and 0.4% of females (varying with ethnicity), with red-green color blindness the most common (Rigden, 1999). Prolific.co asks volunteers to report if they have a color perception deficiency upon registering, and allows experimenters to recruit individuals based on the answer to this question, just like they can select based on age or language (participants do not know this is why they are offered the study). Prolific.co does not ask participants to report what kind of color blindness they have, they were only asked “Do you experience colour blindness?” and given the options: yes, no, rather not say.

Participants (18 to 65) who were fluent in English were recruited from the US/Canada on Prolific.co. First, we recruited 62 participants (the goal was 60 but we

recruited 62 by mistake) who reported color blindness to Prolific.co. Only 4 of these 62 participants were female, a ratio that is close to what we would expect and lending validity to the colorblindness question in Prolific. In wave 2, we recruited 111 participants who did not report colorblindness, 60 of them male. We noted that the average age for colorblind men was higher ($M = 41.4$) than that for non-colorblind men ($M = 34.3$), although it was not clear why. In the next waves, we only recruited men and started with the colorblind group, then recruited non-colorblind men to match the age distribution as best we could by raising the age requirement. The final sample included 117 men in each group, with comparable age (see supplemental material).

Design and procedure.

Participants completed the Survey portion of Studies 1 and 2, including questions about interests, personality and the FNS. Note that we chose not to include two questions about variability in food habits in computing the Food Interest score, so as not to inflate a correlation between Food interest and FNS (see supplemental for discussion).

Results.

Our hypothesis concerns FNS as a function of color blindness. Prior work (Gauthier & Fiestan, 2022; Knaapila et al., 2011; Nezlek & Forestell, 2019) and the survey data from Studies 1 and 2 (in supplemental and summarized in Table 4) indicate correlations between the FNS and other personality traits in the Five Factor Model (Funder, 2001). Therefore, while we had no reason to expect color blindness to be related

to other traits, we report all group differences in Table 3. Reliabilities for all measures are in supplemental materials.

We found a credible difference between groups on FNS in the predicted direction, with color blind men scoring lower relative to non-color-blind men. We also found credible differences in Food interest, Agreeableness, Extraversion, with color blind individuals reporting more interest in food and higher levels of agreeableness and extraversion.

Table 3. Means and differences for Survey variables as a function of group.

	Color blind		Non color blind		Difference	95%CDI diff	effect size (d)	95%CDI effect size
	Mean	SD	Mean	SD				
FNS	25.4	11.7	30	12.5	-4.63	-7.91, -1.39	-0.384	-0.661, -0.109
General Interest	4.91	0.985	4.66	0.895	0.246	-0.003, 0.495	0.261	0.000, 0.532
Food interest	4.39	1.10	4.06	1.01	0.328	0.040, 0.593	0.311	0.051, 0.576
Agreeableness	3.73	0.772	3.42	0.856	0.311	0.094, 0.530	0.382	0.115, 0.658
Conscientiousness	3.62	0.818	3.65	0.785	-0.031	-0.248, 0.176	-0.038	-0.298, 0.229
Extraversion	2.80	0.972	2.51	1.05	0.292	0.034, 0.569	0.289	0.032, 0.563
Imagination	3.95	0.779	3.81	0.815	0.141	-0.073, 0.352	0.177	-0.093, 0.443
Neuroticism	2.7	1.05	2.86	0.924	-0.016	-0.411, 0.098	-0.163	-0.424, 0.103

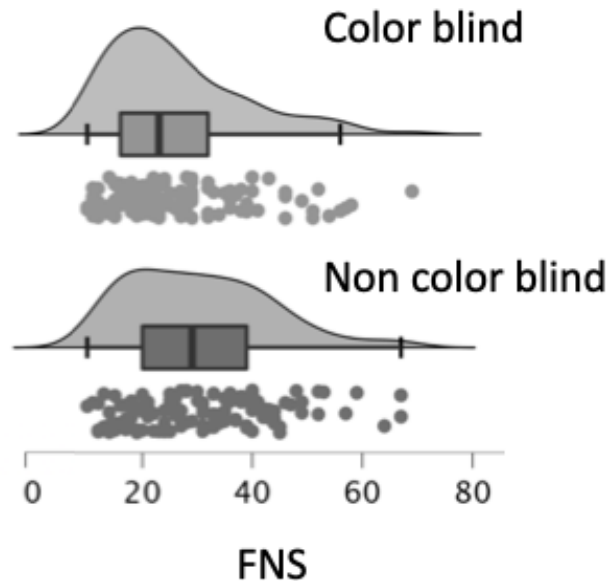


Figure 5. Distributions of FNS scores in color blind and non-color blind men in Study 3.

As summarized in Table 4, these group differences are consistent with reduced food neophobia, given the typical correlations between these variables and FNS in the present work, our Studies 1 and 2, and previous studies (Gauthier & Fiestan, 2022; Nezlek & Forestell, 2019). Those with lower FNS tend to be consistently more interested in food, more agreeable and more extraverted. We found no prior report of differences in personality on the basis of color blindness. In this context, it is difficult to know what to make of the personality differences and their bearing on our hypothesis. Because Agreeableness and Extraversion were both higher in color blind men and correlated with FNS, we tested whether the group difference on FNS survived when controlling for these variables. We found a credible difference in FNS between groups when controlling for Extraversion (mean difference = -3.79, 95% CDI = -6.90, -0.621; Cohen's d = -0.33), and when controlling for Agreeableness (mean difference = -3.38, 95% CDI = -6.56, -0.293;

Cohen's $d = -0.29$), but not when controlling for both Extraversion and Agreeableness, although we also did not find support for the null (mean difference = -2.92 , 95% CDI = $-5.93, 0.176$; Cohen's $d = -0.258$; $BF_{01} = 1.73$).

Table 4. Correlations of Survey and personality variables with FNS, in Study 3, and correlations with FNS from Studies 1 and 2 and other published work.

	S3	G&F22	S1	S2	N&F19
N	234	130	117	77	734
FNS					
General Interest	-.14	-.08	-.13	-.04	
Food interest	-.45	-.53	-.37	-.33	
Agreeableness	-.26	-.31	-.14	-.28	-.14
Conscientiousness	.04	-.12	-.03	-.11	-.06
Extraversion	-.25	-.32	-.08	-.44	-.13
Imagination	-.29	-.21	-.24	-.25	-.21
Neuroticism	.13	.07	-.09	.16	.04

Note: G&F22 is Gauthier & Fiestan (2022). Results from Nezlek & Forestell (2019) calculated from supplementary material. Food interest in other studies than S3 included the two variability questions that could inflate correlation with FNS.

Discussion

Food recognition judgments are impaired by the removal of color. Nevertheless, these judgments can also be performed above chance without color, based on shape and/or texture information. This could simply indicate that color provided some, but not all, of the information diagnostic for food recognition. However, an individual differences approach allowed us to separate domain-general and domain-specific contributions to food recognition performance, and the measurement of both effects survived the removal of color. Our results extend Gauthier and Fiestan's (2022). We find that food-specific recognition ability can be separated into two components: one that is entirely independent of color, responsible for the residual correlations between the two

food recognition tests in every case in this paper, and one that depends on color. This last component, relatively small, appears to be uniquely related to FN.

Figure 6 plots the variance explained by each component in the present work, and for comparison, in Gauthier and Fiestan (2022) who used color in both food tests. A qualitative consideration of these effect sizes tells an interesting story. It is not surprising that the food-specific component is more prominent when both food tests have the same format (both grayscale in Study 2 and both colored in Gauthier & Fiestan, 2022) than when they are different formats in Study 1. In Study 2, the food tests allow no contribution of a domain-general color ability; therefore, o accounted for almost all of the non-domain specific variance in these tests. The domain-general and food-specific contributions combined to account for almost all the variance in performance on the grayscale food tasks (94%) in Study 2.

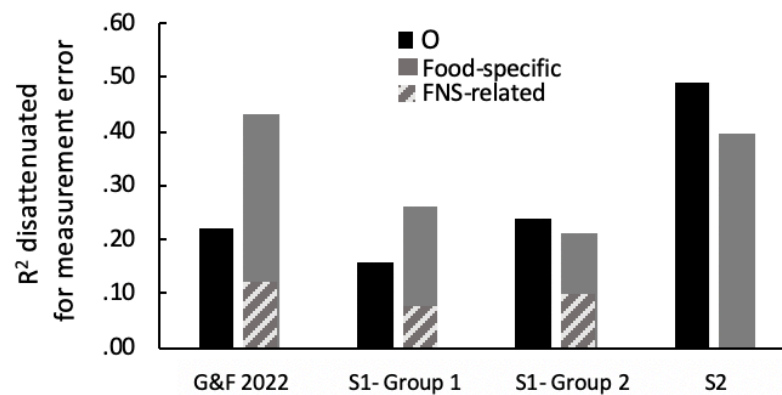


Figure 6. Amount of variance (R^2), disattenuated for measurement error, for each group in Study 1, in Study 2 and for comparison, in Gauthier and Fiestan (2022).

We assume that the portion of the variance in food recognition related to FNS provides an estimate of food-specific variance attributable to color. This accounts for about the same amount of variance across all studies in which at least some food images were in color (12% in Gauthier & Fiestan (2022), 8% and 10% in Study 1). This is part of the domain-specific component in each study. We speculate that *o* accounts for more of the individual differences in Study 2 than other studies because some of the variance in food recognition performance is associated with color processing in the other studies. In contrast, in Study 2, individual differences in food tests could *only* be generated by mechanisms operating in grayscale. Comparing Study 2 to the original work, and because there is no logical reason for a smaller food-specific influence when both food tests were in color than both in grayscale, we estimate that up to 24% of the variance in the Gauthier and Fiestan (2022) study could be attributed to a *domain-general* color ability (Supplemental Table S6). This is obtained by comparing the sum of the variance explained by *o* and a food-specific ability in Study 2 and the original work, $[(40+49) - (22+43)]$. Whether such a *domain-general* color ability is especially relevant to object recognition or depends on individual differences in general color perception remains to be determined.

A completely novel prediction stemmed from our finding a food-specific color ability related to FN: individuals with color blindness should be less food neophobic than those with normal color perception. Our results supported this prediction, although the relationship between this result and effects in Agreeableness and Extraversion remain to be explored. These results are correlational and may be mediated by other variables, but

of all the variables in Study 3, color blindness is most likely to play an antecedent role as it is usually genetically determined and present from birth.

How do our results relate to the ongoing debate about the role of color in food perception? Perhaps most clearly in Study 2, we measured a food-specific recognition ability completely independently of color. While color is used when it is present, most of its contribution appears to be domain-general. The only evidence for a color-dependent food-specific effect is the part of the variance related to FN, the smallest contribution to performance on our food tests, around 10%. It is unclear why FN is specifically related to color. A recent review 30 years of research with the FNS made no mention of color (Rabadán & Bernabéu, 2021). Color can evoke particular emotions (Kaya & Epps, 2004), although the effect depends on context (a pleasant color for a shirt may not be pleasant on a plate, Valdez & Mehrabian, 1994). Judgments of freshness, taste or caloric content can be influenced by cross-modal correspondences between color and taste (Spence, 2019). Whether the role of color in food perception is uniquely tied to affective responses to food should be further explored in future work. In at least one other domain, jury reactions to evidence, removing color from gruesome images reduced emotional reactions without affecting their probative value (Salerno, 2017).

While brain imaging studies have not looked at food-selective responses in grayscale images, our results suggest that at least some of the food-selective ventral pathway areas may remain so without color. This is consistent with the fact that expertise for other categories, novel or familiar (and even a category where color is thought to be important, birds), is related to category-selectivity in the ventral pathway when images are in grayscale or all one color (Gauthier et al., 2000; McGugin et al., 2012; Wong et al.,

2009; Xu, 2005). Color may play a role in ventral pathway object representations, but it has not been found critical for any category. Claims about the visual processing of food should be tested with both color and grayscale images. It is also possible that the portion of the food-specific ability not related to color and affective responses could be more general than we suggested here, and also apply to more complex “scene-like” objects other than food. While *o* can account for recognition of complex familiar objects on natural backgrounds (Richler et al., 2019; Sunday et al., 2018), and other complex stimuli like radiographs or blood cells assays (Smithson et al., 2022; Sunday et al., 2017), it may not control for all domain-general information in food images.

Several studies have been concerned with a food/nonfood distinction (e.g., Kumar et al., 2016; Tsourides et al., 2016), a task that may rely to some extent on different dimensions than the more specific discriminations we measured here. This distinction is often argued to be “critical to our survival” (Tsourides et al., 2016, p. 1). But nowadays, decisions about what to eat are likely more important than food/non-food judgments. In our modern world, food is easily located in kitchens, stores, and represented on menus. Those who lack in food rarely do so because they cannot find it, but because they cannot afford it. Specialized and innate mechanisms for food detection could have been shaped by pressure to separate food from non-food (Polyak, 1957). The finding that color facilitates the rapid detection of food (Sato, 2021) has been invoked as support for this theory, notwithstanding the challenges of testing specific evolutionary explanations (Confer et al., 2010). While food detection fits the bill for a function with fitness relevance, a trait relevant for subordinate-level food judgments could also have experienced natural selection, but could also be learned. Measuring the recognition of

individual dishes as we have done may be most helpful in accounting for neural responses to food images in adults who have experienced a lifetime of complex visual images related to food. It may be less helpful in determining the evolutionary origins of this ability.

Our results reveal a food recognition ability independent of color and a small, but replicable, color-specific food recognition ability related to FN. This work illustrates how an individual differences approach can facilitate the study of visual mechanisms relevant to everyday decisions.

Author contributions

J. Sun and I. Gauthier designed and ran the studies, analyzed data and prepared the manuscript.

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